EDA playground

New Run Save*

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Doulos does not endorse training material from other suppliers on EDA Playground.

Languages & Libraries

Testbench + Design

C++/SystemC

Libraries

None

SystemC 2.3.3

Tools & Simulators

Examples

236

testbench.cpp

```
#include <systemc>
#include <tlm>
#include <tlm_utils/simple_initiator_socket.h>
#include <tlm_utils/simple_target_socket.h>
using namespace sc_core;
using namespace tlm;
using namespace std;
// Target
struct Target : sc_module {
    tlm_utils::simple_target_socket<Target> socket;
    SC_CTOR(Target) : socket("socket") {
        socket.register_b_transport(this,
&Target::b_transport);
    }
    void b_transport(tlm_generic_payload& trans, sc_time&
delay) {
        int* data = (int*) trans.get_data_ptr();
        cout << "Target received: " << *data << endl;
        delay += sc_time(10, SC_NS);
    }
};
// Initiator
struct Initiator : sc_module {
    tlm_utils::simple_initiator_socket<Initiator> socket;
    SC_CTOR(Initiator) : socket("socket") {
        SC_THREAD(thread_process);
    }
    void thread_process() {
        tlm_generic_payload trans;
        sc_time delay = SC_ZERO_TIME;
        int data = 555;
        trans.set_data_ptr(reinterpret_cast<unsigned char*>
(&data));
        trans.set_data_length(sizeof(data));
        cout << "Initiator sending: " << data << endl;
        socket->b_transport(trans, delay);
        wait(delay);
    }
};
int sc_main(int argc, char* argv[]) {
    Initiator initiator("initiator");
    Target target("target");
    initiator.socket.bind(target.socket);
    sc_start();
    return 0;
}
```

Log Share

```
[2026-11-05 03:21:34 UTC] g++ -o sim -lsystemc *.cpp  && echo "Compile done. Starting run..." && ./sim
Compile done. Starting run...
Initiator sending: 555
Target received: 555

        SystemC 2.3.3-Accellera --- May 23 2020 14:33:56
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Done
```